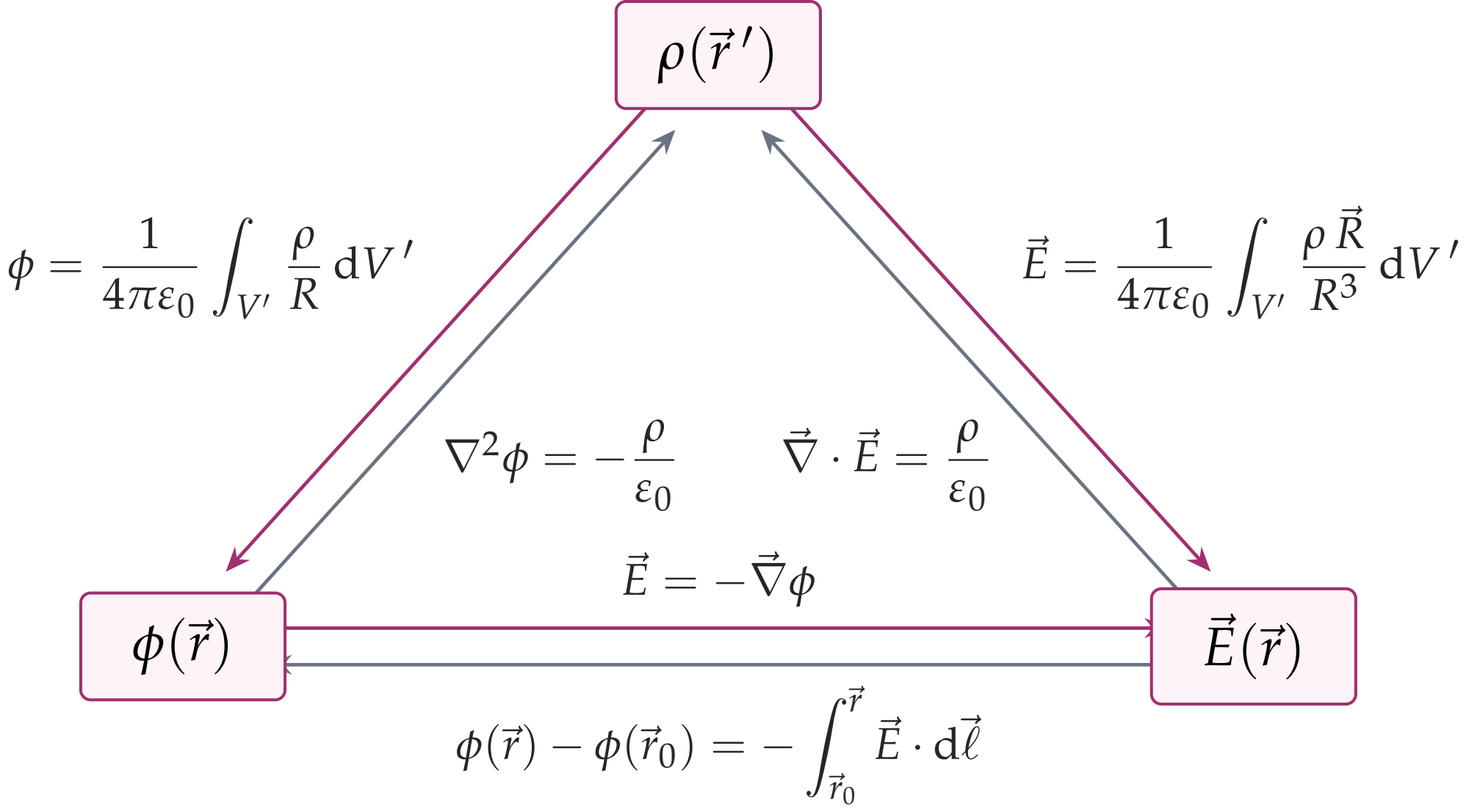

$\rho(\vec{r}\,')$
$\phi = \frac{1}{4\pi\varepsilon_0}\int_{V'}\frac{\rho}{R}\,\mathrm{d}V'$
$\vec{E} = \frac{1}{4\pi\varepsilon_0}\int_{V'}\frac{\rho\,\vec{R}}{R^3}\,\mathrm{d}V'$
$\nabla^2\phi = -\frac{\rho}{\varepsilon_0}$
$\vec{\nabla}\cdot\vec{E} = \frac{\rho}{\varepsilon_0}$
$\vec{E} = -\vec{\nabla}\phi$
$\phi(\vec{r})$
$\vec{E}(\vec{r})$
$\phi(\vec{r}) - \phi(\vec{r}_0) = -\int_{\vec{r}_0}^{\vec{r}}\vec{E}\cdot\mathrm{d}\vec{\ell}$